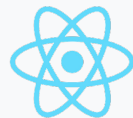


DataStax

Developers



React



GraphQL

Building a **NetFlix** clone

using GraphQL, React and JAMStack



01



Housekeeping

Live and Hands-on

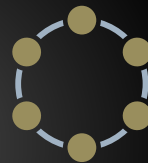
02



Our Goal

Component overview

03



Database Setup

Data model & Astra DB

04



GraphQL

GraphQL

Learn & Hands-On

05



Serverless + Client

Run the application

06



What's next?

Quiz, Homework, ...



Agenda





Housekeeping

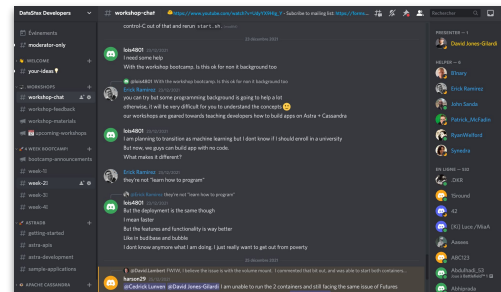
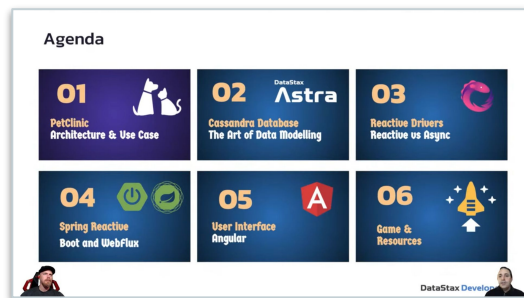
Live and Hands-on



Livestream: youtube.com/DataStaxDevs

Questions: <https://dtsx.io/discord>

Live and interactive



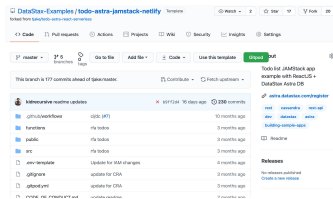
Games and quizzes: menti.com



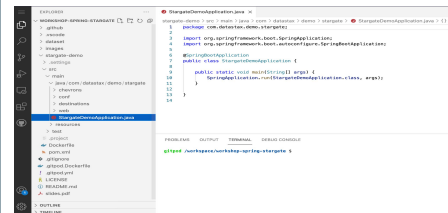
Live Sessions

Nothing to install!

Source code, exercises, slides



IDE

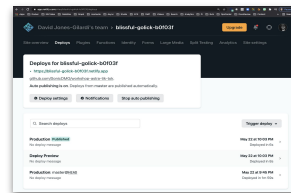


Database + GraphQL + PlayGround



DataStax
Astra DB

Deploy + Run + Host



Hands-On Housekeeping



Get your badge for today

- **BEGINNERS**

- Relax, no complex code to write
- Get the most of it by following what presenters do and ask Questions !
- Try to improve and develop the application later!

- **INTERMEDIATE**

- Scroll and do exercises on your own PACE
- Sample code, ready to go to copy-paste

- **EXPERTS**

- Run stuff locally
- Challenge



Choose your path



01



Housekeeping

Live and Hands-on

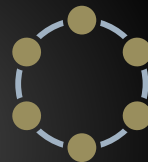
02



Our Goal

Overall architecture

03



Database Setup

Data model & Astra DB

04

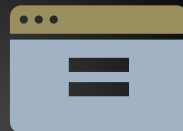


GraphQL

GraphQL

Learn & Hands-On

05



Serverless + Client

Run the application

06



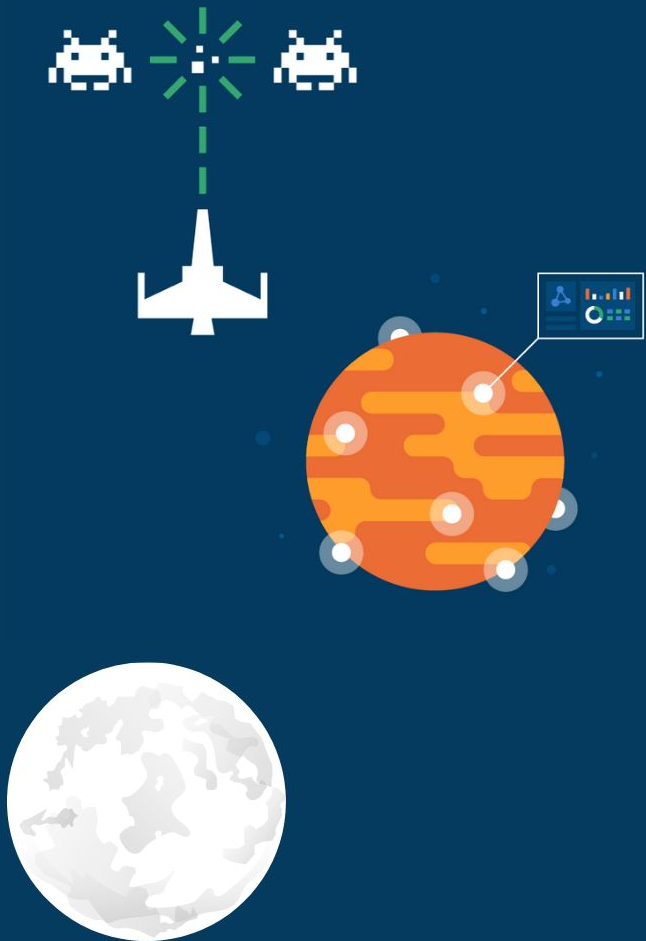
What's next?

Quiz, Homework, ...



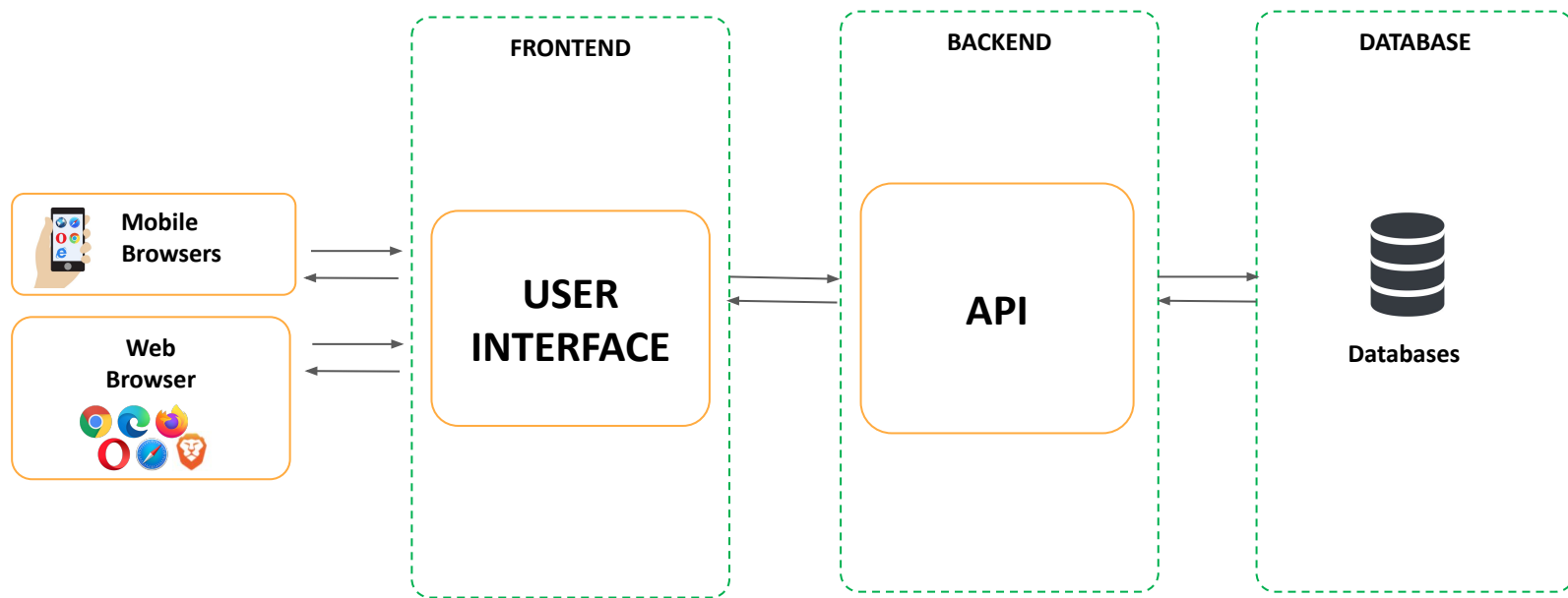
Agenda



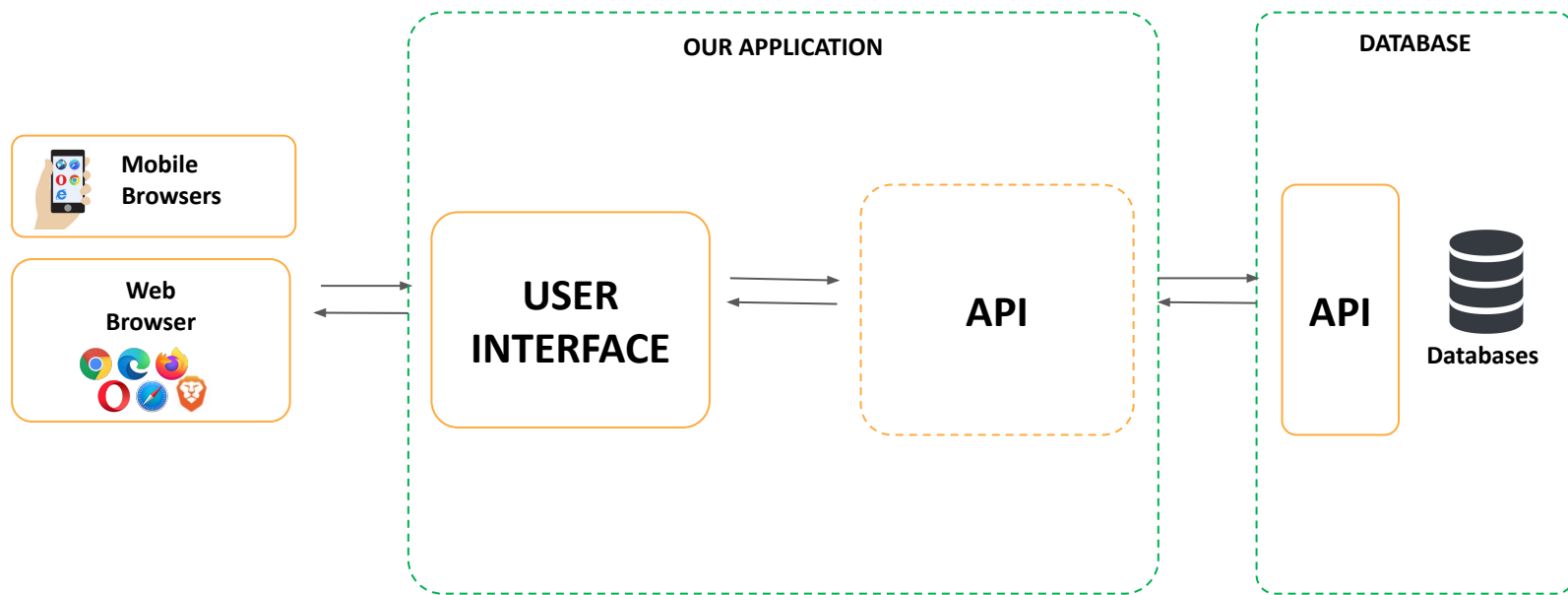


Our Goal

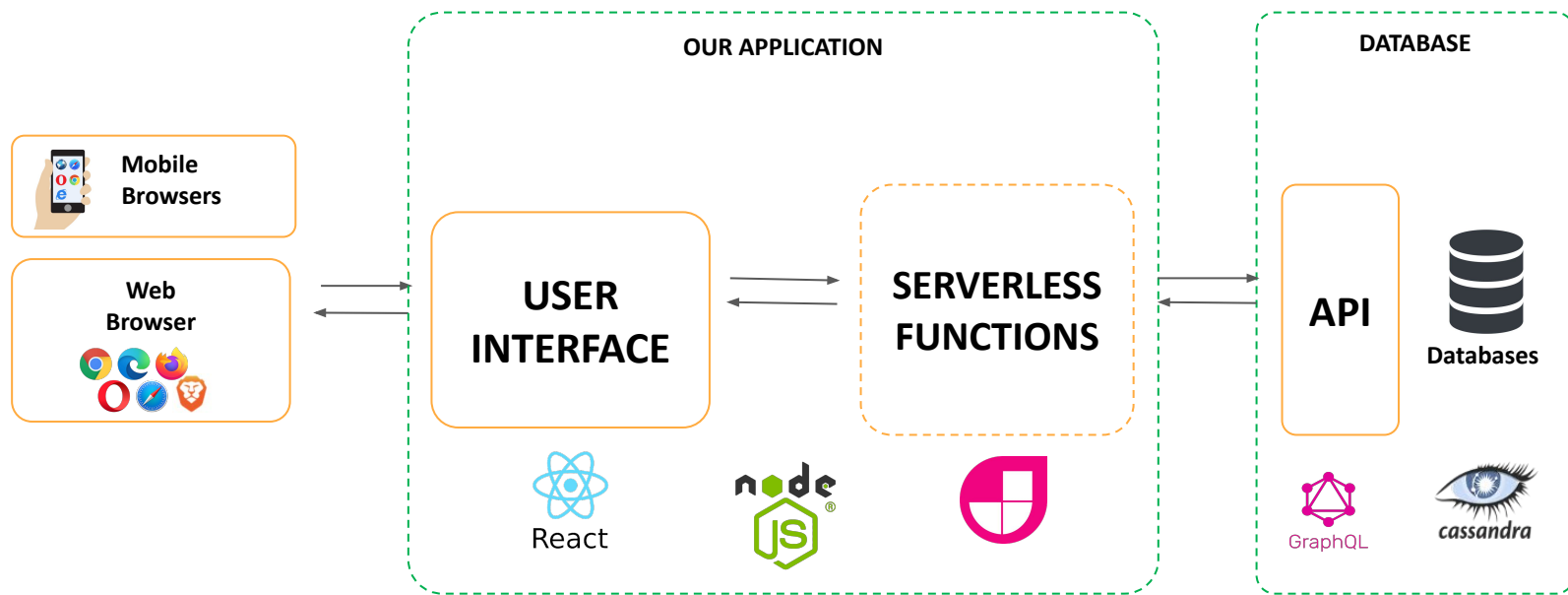
Overall Architecture



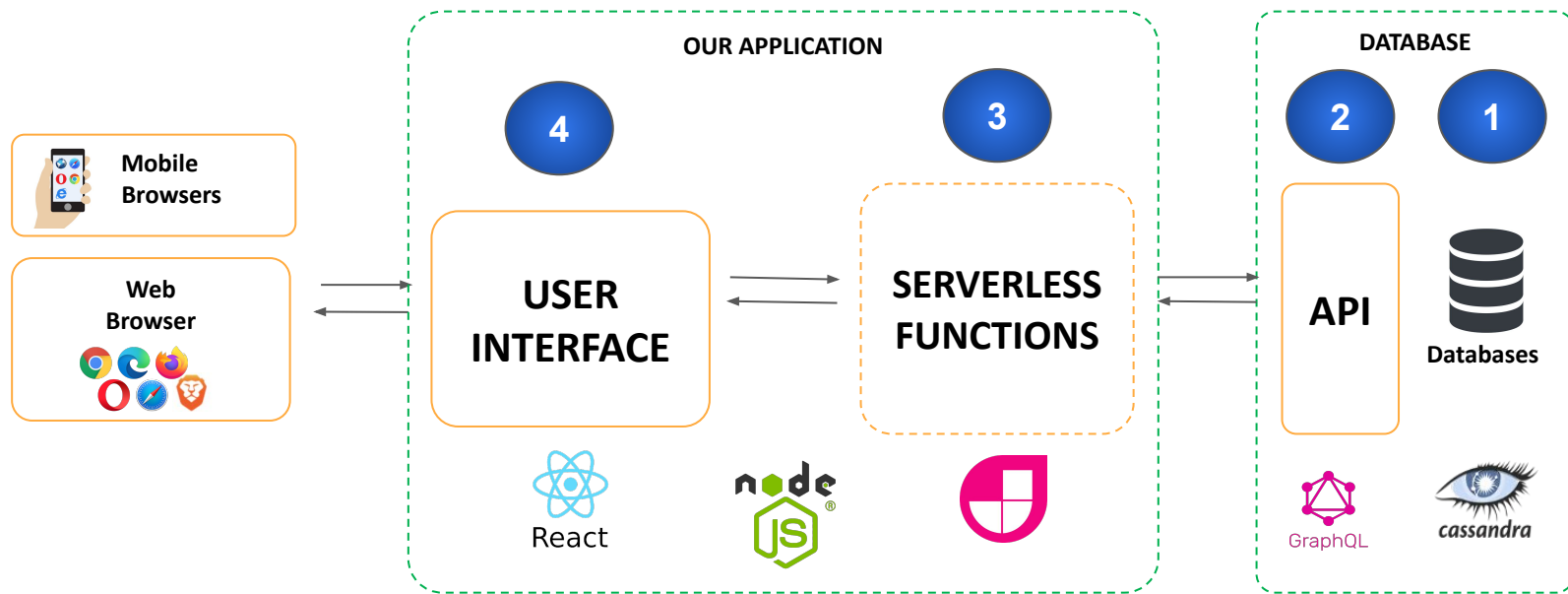
Building a Full-stack application



Building a Full-stack application



Building a Full-stack application



Gitpod



netlify

DataStax
Astra DB

Agenda

01



Housekeeping

Live and Hands-on

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Database Setup

Data model & Astra DB

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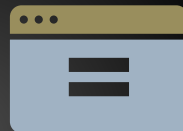


GraphQL

GraphQL

Learn & Hands-On

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Serverless + Client

Run the application

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What's next?

Quiz, Homework, ...



Agenda

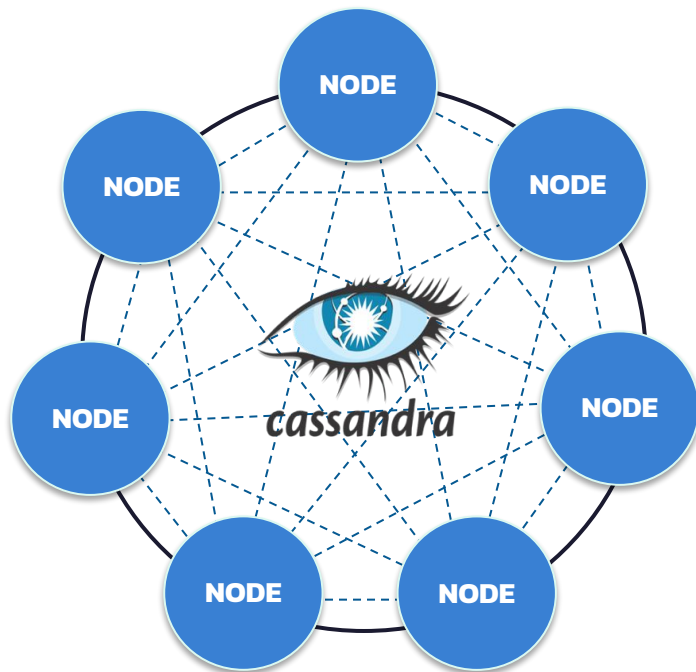


Database setup



Data model & Astra DB





- Big Data Ready
- Read / Write Performance
- Linear Scalability
- Highest Availability
- Self-Healing and Automation
- Geographical Distribution
- Platform Agnostic
- Vendor Independent



Introduction to Apache Cassandra™



CQL API
Compatibility
It is Apache Cassandra !

PULSAR CLIENT
Compatibility
It is Apache Pulsar !

GRAPHQL API
Discoverability, Flexibility
Read only what you need

DOCUMENT API
Flexible Schema
Document-Oriented usages

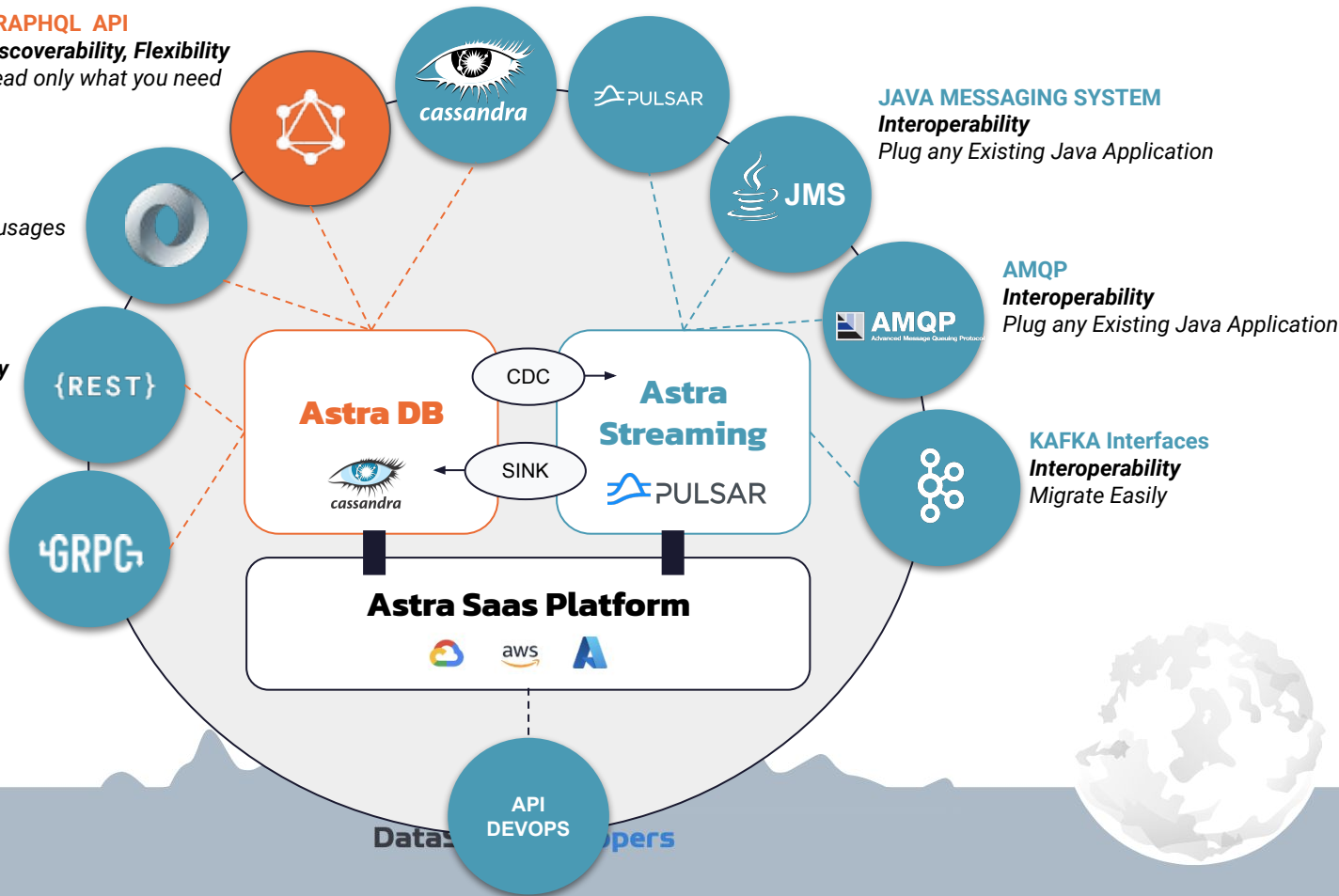
REST API
Stateless, Interoperability
CRUD for Cassandra

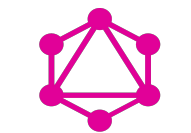
GRPC API
Performance, Interoperability
Execute cql in the fastest way

JAVA MESSAGING SYSTEM
Interoperability
Plug any Existing Java Application

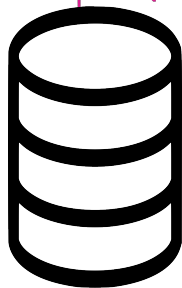
AMQP
Interoperability
Plug any Existing Java Application

KAFKA Interfaces
Interoperability
Migrate Easily





GraphQL



Astra DB

Get the movie genres

get (a pageful of) reference items labeled 'genre'

Get movies for a genre

get (a pageful of) movies for a given genre

Data modeling, the Cassandra way:

"design the table after the query"

(also: tables are partitioned!)



DB usage patterns





reference_list		
label	text	K
value	text	C↑

movies_by_genre		
genre	text	K
year	int	C↓
title	text	C↑
duration	int	
synopsis	text	
thumbnail	text	



get (a pageful of) reference items labeled 'genre'

get (a pageful of) movies for a given genre

"partition key" (rows stored together)

"primary key" (row uniqueness)

"Chebotko diagrams"

label	value
genre	Action
genre	Anime
genre	Award-Winning
genre	Children & Family
...	

reference_list		
label	text	K
value	text	C↑

genre	year	title	duration
Comedies	2020	AJ and the Queen	142
Comedies	2020	All the Freckles in the World	148
Comedies	2020	Leslie Jones: Time Machine	140
Comedies	2020	Live Twice, Love Once	123
Comedies	2019	"Gabriel "Fluffy" Iglesias: One Show Fits All"	134
...			
Reality TV	2019	Fastest Car	117
Reality TV	2019	Flinch	111
Reality TV	2019	Haunted	124
Reality TV	2019	Hyperdrive	144
Reality TV	2019	I'm with the Band: Nasty Cherry	119
Reality TV	2019	Jailbirds	140
...			

movies_by_genre		
genre	text	K
year	int	C↓
title	text	C↑
duration	int	
synopsis	text	
thumbnail	text	

stored together)

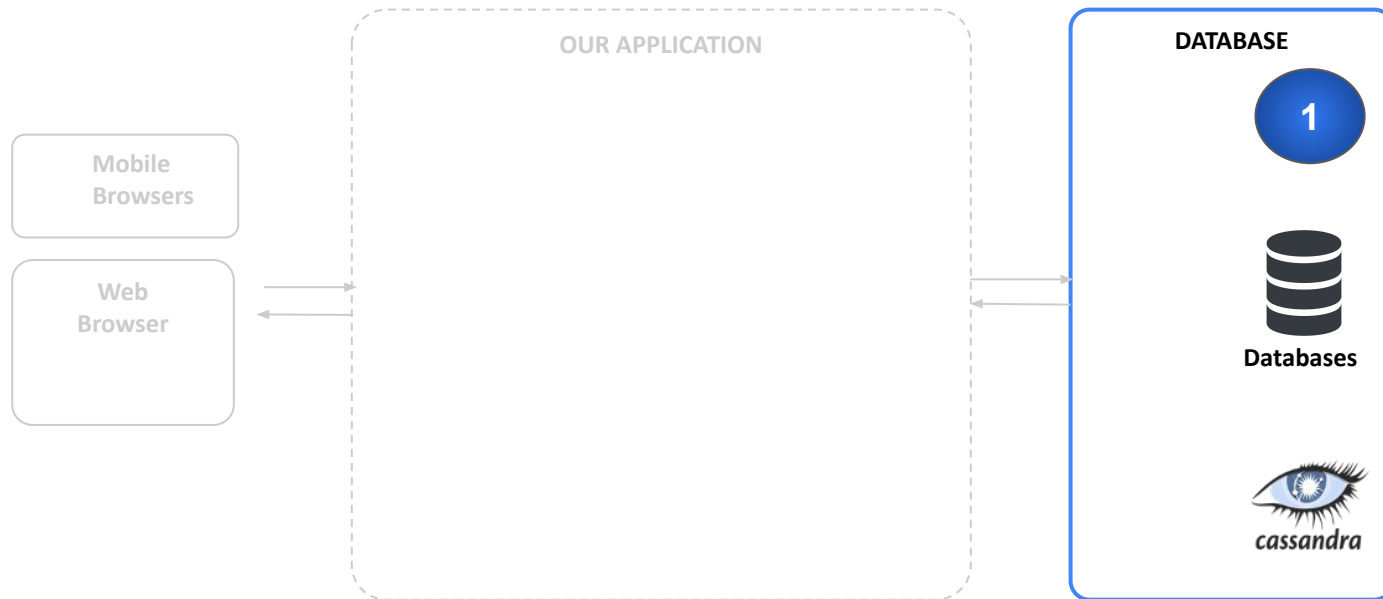
"primary key" (row uniqueness)

"Chebotko diagrams"

Table row examples

DataStax

Astra DB



Create your Astra DB

Github/datastaxdevs



Create Astra DB

DataStax

Astra DB

OUR APPLICATION

Mobile
Browsers

Web
Browser

DATABASE

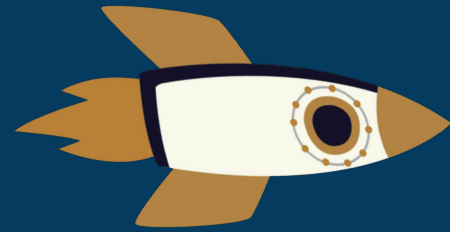
1



Databases



Create your Astra DB



Hands-on (!github)

DB SETUP

- ✓ Create (or resume) your Database
- ✓ Create a token



01



Housekeeping

Live and Hands-on

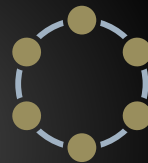
02



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Overall architecture

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Database Setup

Data model & Astra DB

04

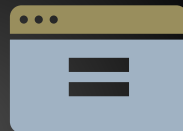


GraphQL

GraphQL

Learn & Hands-On

05



Serverless + Client

Run the application

06



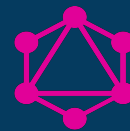
What's next?

Quiz, Homework, ...



Agenda





GraphQL

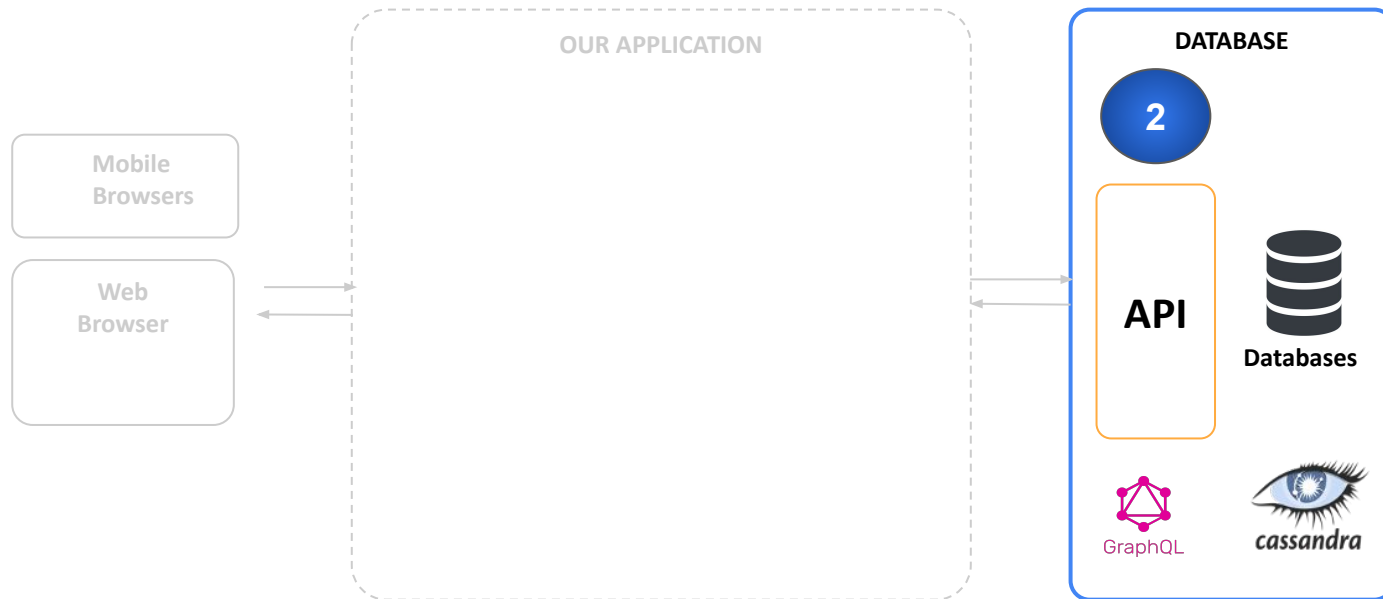
GraphQL

Learn & Hands-on



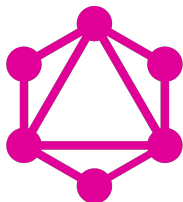
DataStax

Astra DB



GraphQL to access data

- 2012 Created by Facebook
Used internally for mobile apps
- 2015 Facebook give talk at ReactJs Conf
- 2015 Facebook open sourced graphQL
- 2016 Github announced move to GQL



GraphQL

Definition

GraphQL is an application programming interface (API) query language and server-side runtime that prioritises giving customers precisely the data they request.

Meet GraphQL



Describe your data

```
type Project {  
  name: String  
  tagline: String  
  contributors: [User]  
}
```

Ask for what you want

```
{  
  project(name: "GraphQL") {  
    tagline  
  }  
}
```

Get predictable results

```
{  
  "project": {  
    "tagline": "A query language for APIs"  
  }  
}
```



Why GraphQL ?





- ❖ Discoverability, introspection
- ❖ Declarative Data Fetching
- ❖ Schema stitching
- ❖ Customize responses
- ❖ Flexible versioning
- ❖ Match standards (Json | Http)



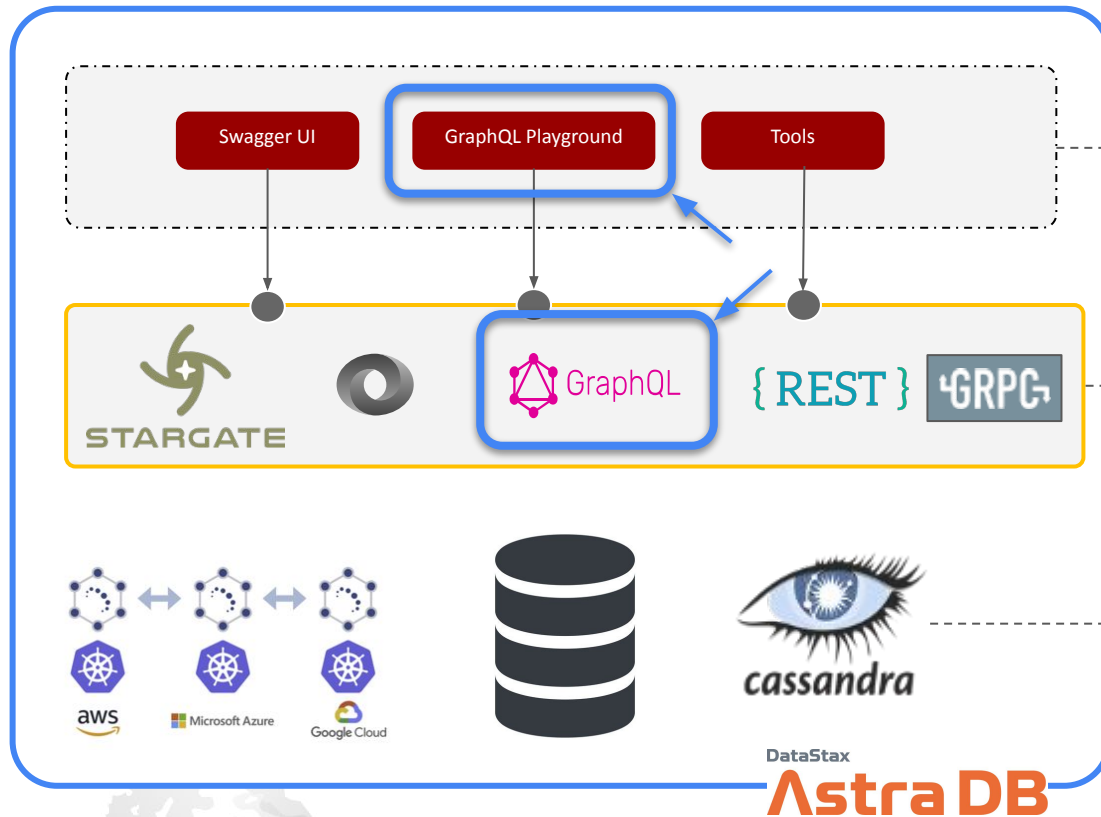
- ❖ Single endpoint (*versioning, monitoring, security*)
- ❖ Complex implementation (*tooling, still young*)
- ❖ Nice for customers, nasty for DB (*N+1 select*)
- ❖ Rate limiting is required
- ❖ Caching is hard (Apollo is a good solution here <https://www.apollographql.com>) <- this is what Netflix uses



- ❖ BFF : Backend for frontend
- ❖ Service aggregation | composition, "federation" (*joins*)
- ❖ When bandwidth matters (*mobile phones*)



GraphQL S.W.O.T. analysis



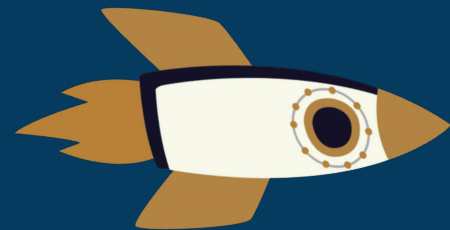
User Interface
Web Utils

OSS Stargate.io
Gateway

Apache Cassandra



Astra DB's GraphQL layer



Hands-on (!github)

DATA SETUP

- ✓ Schema with GraphQL Playground
- ✓ Read/write data with Playground
- ✓ Bulk-import data into DB

01



Housekeeping

Live and Hands-on

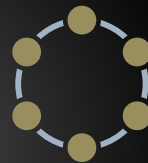
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GraphQL

Learn & Hands-On

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Serverless + Client

Run the application

06



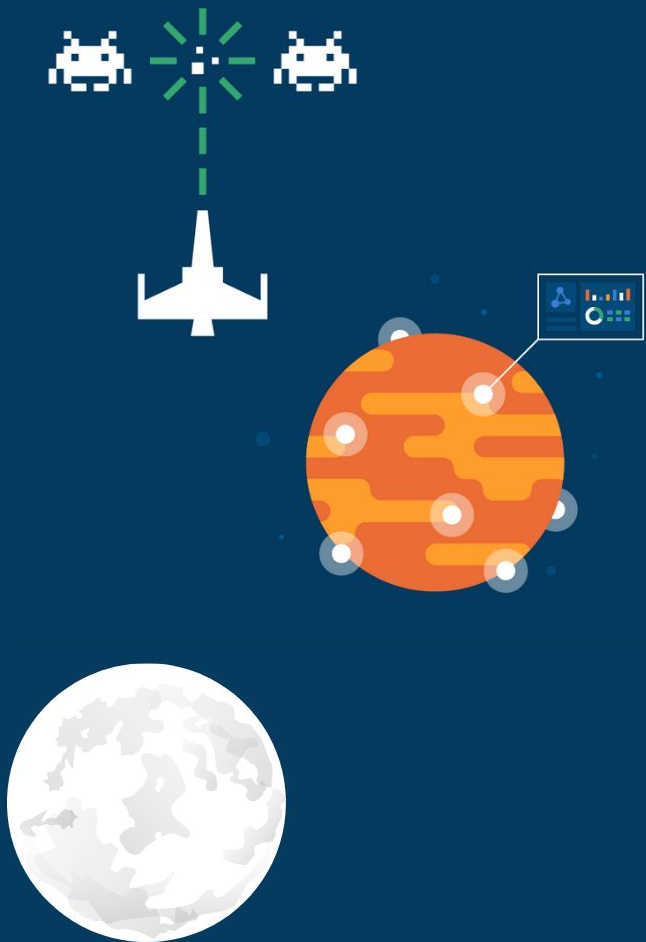
What's next?

Quiz, Homework, ...



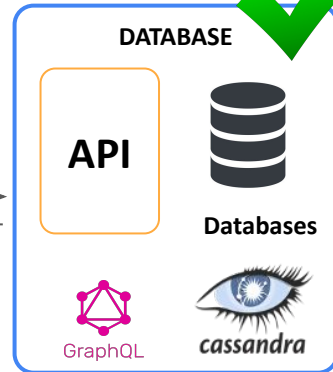
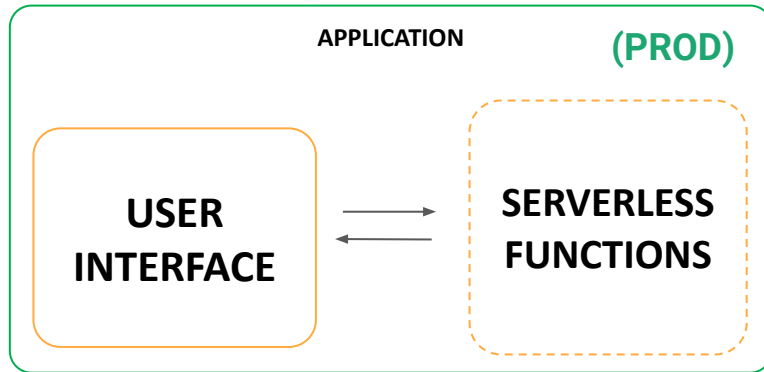
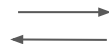
Agenda





Serverless + client

Run the application

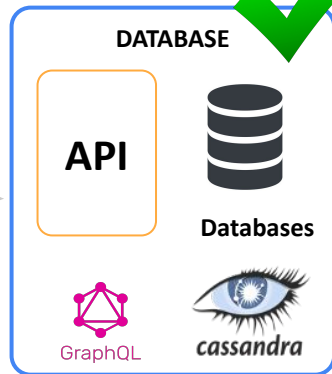
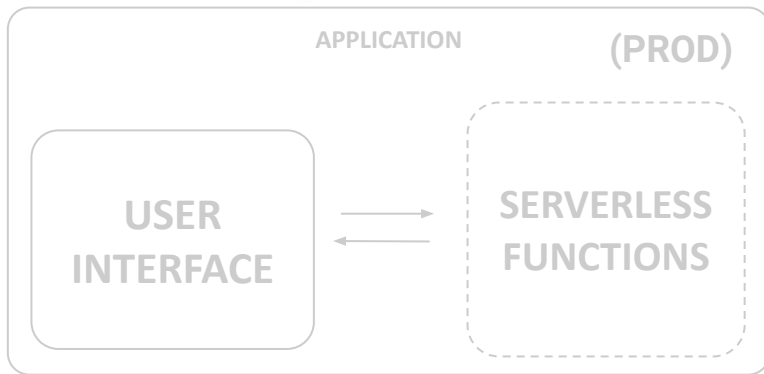


Desired prod result

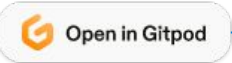


DataStax

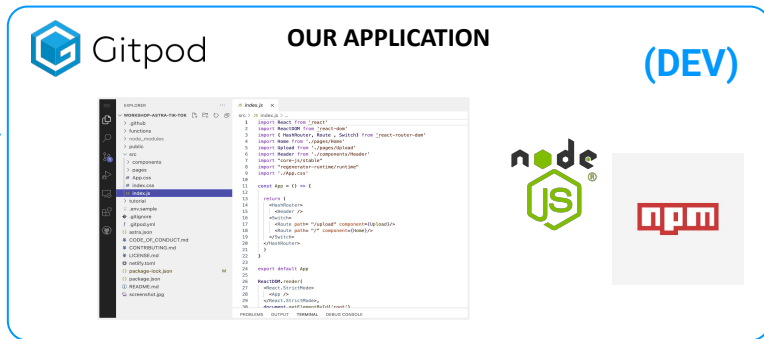
Astra DB



Code + Setup + Run Locally

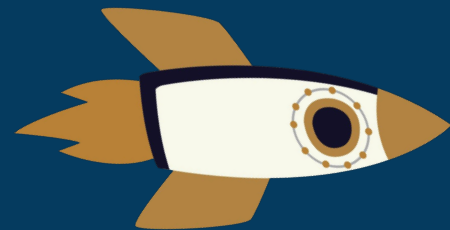


Github/datastaxdevs



"just" dev run on Gitpod

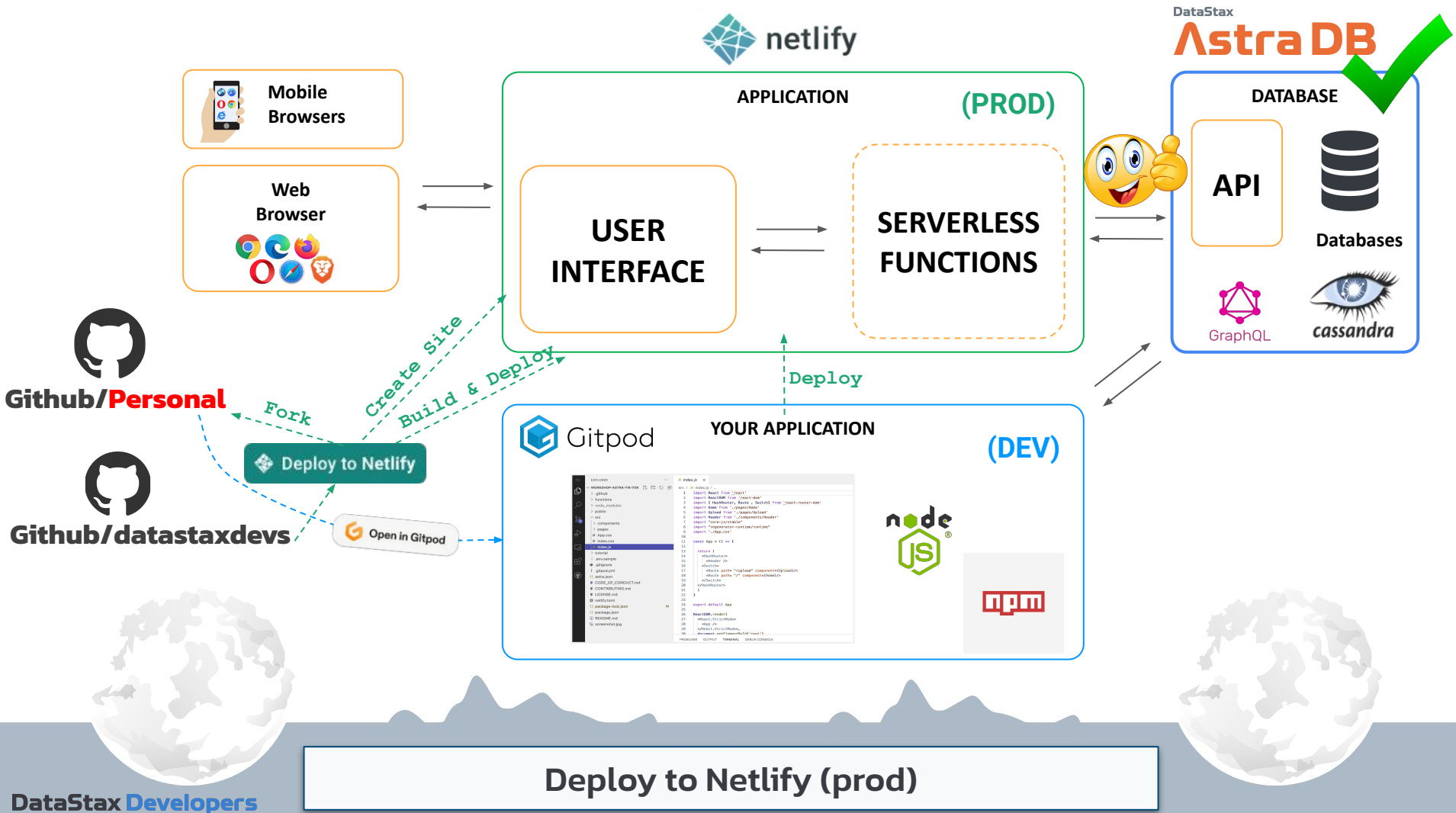
DataStax Developers



Hands-on (!github)

RUN APP

- ✓ Run app on Gitpod (dev mode)
- ✓ Inspect serverless + client code
- ✓ *(Deploy in prod to Netlify)*



01



Housekeeping

Live and Hands-on

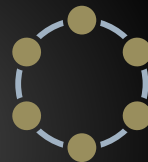
02



Our Goal

Overall architecture

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Database Setup

Data model & Astra DB

04

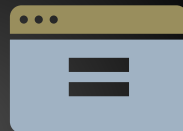


GraphQL

GraphQL

Learn & Hands-On

05



Serverless + Client

Run the application

06



What's next?

Quiz, Homework, ...



Agenda



What's next?



Quiz, Homework, ...



- **Homework**

- Complete hands-on by adding YOUR OWN movie to the DB
- Show the NEW movie in the running app (Github is fine)
- Submit screenshot to us ... and wait for your **badge**!

- **Optionally**

- Deploy to Netlify!
- Client code deep dive with Ania's video

- Do reach us for anything: dtsx.io/discord !

All info on Github - !homework



Assignment and Badge



DataStax Developers | #workshop-chat | <https://www.youtube.com/watch?v=MuwT5xkFVWI> - Subscribe to mailing list: [http...](http://...)

Left Sidebar (Channels):

- Événements
- #moderator-only
- WELCOME
- start-here
- code-of-conduct
- introductions
- upcoming-events
- useful-resources
- memes
- your-ideas
- the-stage
- WORKSHOPS
- #workshop-chat (selected)
- workshop-feedback
- workshop-materials
- upcoming-workshops
- ASTRADB
- getting-started
- astra-api
- astra-development
- sample-applications
- APACHE CASSANDRA

Channel Content:

RIGGITYREKT Hier à 21:14
I have a 5 node datacenter, 4 nodes are on dse version 5.1.20, one is on dse5.0.15. I am doing some mixed version testing for a class and the one node that is 5.0.15 is coming up as an analytics workload. I dont have /etc/default/dse, instead I am using /etc/init.d/dse-cassandra.
how do i make that node start in cassandra workload, not in analytics?

RIGGITYREKT Hier à 23:39
Okay I found out my issue, when i started DSE 5.0.15 it had endpointsnitch set to DseSimpleSnitch, the rest of my cluster is using PropertyFileSnitch, when i change it to PropertyFileSnitch, it still uses the simple snitch config. looking at the docs i see there is a way to go to GossipingPropertyFileSnitch, but i need the property file one. I can wipe this dbs, do anything with this node to get this done. how do i fix this?
[@here](#)

19 novembre 2021

@RIGGITYREKT Okay I found out my issue, when i started DSE 5.0.15 it had endpointsnitch set to DseSimpleSnitch, the rest...
Erick Ramirez Aujourd'hui à 02:19
mixed versions isn't supported and you're guaranteed to run into weird issues that will cause further problems down the track

@RIGGITYREKT I have a 5 node datacenter, 4 nodes are on dse version 5.1.20, one is on dse5.0.15. I am doing some mixed v...
Cedrick Lunven Aujourd'hui à 09:01
When you start a node you have parameters -k for analytics, -g for graph and -s for search. To remove analytics check and remove -k

Right Sidebar (List of Members):

PRESENTER — 1

- David Jones-Gilardi

HELPER — 7

- 012345
- AaronP
- BINARY
- Chelsea Navo
- Jeremy Hanna
- John Sanda
- Patrick_McFadin

EN LIGNE — 560

- samu-
- 6304-42J8
- Aahlya
- Abdurahim
- abhi3pathi
- Abhiis.s
- Abhineet
- Abirsh

Bottom Bar: Envoyer un message dans #workshop-chat

!discord

dtsx.io/discord

Datastax Developers Discord (18k+)

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to Cassandra for Developers
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Databases
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Build your own NETFLIX
clone!
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Streamed 2 weeks ago



Build your own NETFLIX
clone!
7.4K views •
Streamed 2 weeks ago



Astra Streaming Demo
177 views • 2 weeks ago



Kubernetes Ingress
Management with Traefik...
496 views •
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Build your own TikTok clone!
1.9K views •
Streamed 3 weeks ago



Build your own TikTok
Clone!
4K views •
Streamed 3 weeks ago



How to use the Connect
Driver in Astra DB
113 views • 4 weeks ago



How to use the CQL Console
in Astra DB
39 views • 4 weeks ago



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Authentication Token in...
37 views • 4 weeks ago



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Gallery
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in Astra DB
42 views • 4 weeks ago



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Room (Workshops)
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Streamed 4 weeks ago



Cassandra Day India: RF
Room (Talks)
1.3K views •
Streamed 1 month ago

Thank You!

